

Test Case Design – Task

Management Application

Since the application is close to production and there is no existing test documentation, testing should cover **functional, negative, edge, validation, security, and database-related cases**.

1. Registration Test Cases

Test Case ID	Test Case	Expected Result
REG-01	Register with correct name, email and password	User should be registered successfully
REG-02	Register using an email which is already registered	Error message should be shown
REG-03	Enter wrong email format	It should show an invalid email message
REG-04	Leave all fields blank	Required field messages should be shown
REG-05	Leave any one required field blank	Registration should not be completed
REG-06	Enter a short password	Password validation message should appear
REG-07	Enter different password and confirm password	Error should be displayed
REG-08	Register with valid details	Account should be created
REG-09	Enter very long name/email	Application should handle it properly
REG-10	Enter special characters in the fields	Application should not crash

2. Login Test Cases

Test Case ID	Test Case	Expected Result
LOG-01	Login with correct email and password	User should login successfully
LOG-02	Login with wrong password	Error message should be displayed
LOG-03	Login with an unregistered email	Login should fail
LOG-04	Keep email and password empty	Required field message should appear
LOG-05	Enter invalid email	Validation message should appear
LOG-06	Enter wrong password in different case	Login should not be successful
LOG-07	Enter very long login details	Application should handle the input properly
LOG-08	Enter SQL type input in login fields	Login should fail safely
LOG-09	Refresh page after login	User should remain logged in if session is valid
LOG-10	Logout and press browser back button	User should not access the account again

3. Task CRUD Test Cases

Create Task

Test Case ID	Test Case	Expected Result
C-01	Create a task with valid details	Task should be created
C-02	Create task with empty title	Error message should be shown

Test Case ID	Test Case	Expected Result
C-03	Create task with valid title and description	Task should be saved in database
C-04	Create more than one task	All tasks should be created
C-05	Enter a very long task title	Application should handle it properly

View Task

Test Case ID	Test Case	Expected Result
V-01	Open task list after login	User's tasks should be displayed
V-02	User has no tasks	Empty task message should be displayed
V-03	Create a task and refresh the page	Task should still be present
V-04	Create multiple tasks	All created tasks should be visible
V-05	Login with another user	Previous user's tasks should not be visible

Edit Task

Test Case ID	Test Case	Expected Result
E-01	Edit an existing task	Task should be updated
E-02	Change task title	New title should be displayed
E-03	Remove required task information	Validation message should appear
E-04	Cancel editing	Old task information should remain
E-05	Edit task and refresh page	Updated information should remain

Delete Task

Test Case ID	Test Case	Expected Result
D-01	Delete a task	Task should be deleted
D-02	Delete task and refresh page	Deleted task should not appear
D-03	Cancel delete operation	Task should remain
D-04	Delete all tasks	Empty task list should be displayed
D-05	Try to delete another user's task	User should not be allowed to delete it

4. Input Validation and Error Handling

Test Case ID	Test Case	Expected Result
VAL-01	Submit form without entering data	Proper error messages should appear
VAL-02	Enter only spaces in a field	It should not accept invalid input
VAL-03	Enter special characters	Application should handle them properly
VAL-04	Enter HTML/script code	Code should not execute
VAL-05	Enter SQL injection input	Application should remain safe
VAL-06	Database is not available	Proper error message should be shown
VAL-07	Click Create button multiple times quickly	Duplicate tasks should not be created
VAL-08	Internet/server connection fails while saving	User should be informed that saving failed
VAL-09	Session expires	User should be asked to login again
VAL-10	Unexpected error occurs	Application should show a simple error message

5. Edge Cases

Some extra cases which should also be checked:

- Register with the same email using different capital letters.
- Create a task with a very long title.
- Create a task with only spaces.
- Try to create a task without logging in.
- Refresh the page after creating a task.
- Click the delete button multiple times.
- Open the same task in two browser tabs.
- Edit a task from two tabs at the same time.
- Try to access another user's task manually.
- Try to use the application when the database is temporarily unavailable.

Bug Identification

Bug ID	Possible Bug	Severity	Reason / Impact
BUG-01	One user can see another user's tasks	Critical	User's private information can be accessed
BUG-02	SQL injection is possible in login or task fields	Critical	Database information may be stolen or changed
BUG-03	Passwords are stored without proper encryption/hashing	Critical	Passwords can be exposed if database is leaked
BUG-04	Task gets deleted without asking for confirmation	Major	User may delete a task accidentally
BUG-05	Task disappears after refreshing the page	Major	Task may not have been saved correctly
BUG-06	Clicking Create many times creates duplicate tasks	Major	Unnecessary duplicate data will be created
BUG-07	Application accepts invalid input	Major	Bad or unwanted data can be stored

Bug ID	Possible Bug	Severity	Reason / Impact
BUG-08	User can access the account after logout using Back button	Critical	Other people may access the user's account
BUG-09	Script entered in task details gets executed	Major	This can create a security problem
BUG-10	Technical/database error is shown directly to the user	Minor/Major	It looks unprofessional and may expose system information

Conclusion

Before releasing the application, I would first test **registration and login**, then check **user security and task CRUD operations**. After that, I would test **validation, database saving, error handling and edge cases**. This will help find major problems before the application is released.